

Day	Phase	Topic	What to Learn / Build
1	Phase 1	What is an Agent?	Understand agent vs prompt vs workflow. Read Anthropic's agent overview.
2	Phase 1	Planning & Execution	Learn how agents plan steps vs execute them. Study ReAct pattern basics.
3	Phase 1	Tool Use & Memory	Understand tool calling, short-term vs long-term memory concepts.
4	Phase 1	Build: Simple Agent	Build input → decide → call tool → return result. Use a basic calculator tool.
5	Phase 2	Function / Tool Calling	Deep dive into function calling with structured inputs and outputs.
6	Phase 2	Structured Outputs	Use Pydantic / JSON schema to enforce structured agent outputs.
7	Phase 2	Error Handling & Retries	Handle tool failures, add retry logic, graceful fallbacks.
8	Phase 2	Build: Tool-Using Agent	Build agent with 3 tools: Calculator, Web Search, Database Query.
9	Phase 3	Agent Roles	Study Planner, Executor, Critic roles and when to use each.
10	Phase 3	Sequential Communication	Build a simple sequential pipeline — Agent A passes output to Agent B.
11	Phase 3	Hierarchical (Supervisor)	Implement a Supervisor agent that delegates to sub-agents.
12	Phase 3	Collaborative Patterns	Study collaborative multi-agent patterns where agents share context.
13	Phase 3	Assignment 1 – Setup	Set up project. Implement Supervisor + Researcher agents. Wire tool calls.
14	Phase 3	Assignment 1 – Build	Implement Analyst + Writer agents. Connect full Researcher→Analyst→Writer flow.
15	Phase 3	Assignment 1 – Polish	Test end-to-end. Generate blog: 'Agent as Tool vs Sub-Agent'. Review output.
16	Phase 4	RAG Fundamentals	Understand Retrieval-Augmented Generation — chunking, embedding, retrieval.
17	Phase 4	Vector Databases	Set up a vector DB (Chroma / FAISS / Pinecone). Store and query embeddings.
18	Phase 4	Memory in Agents	When to use memory vs not. Implement short-term (buffer) + long-term (vector) memory.
19	Phase 4	Assignment 2 – Setup	Set up debate chatbot project. Implement Moderator, Pro Agent, Con Agent.
20	Phase 4	Assignment 2 – Debate Flow	Wire debate rounds: Opening → Rebuttal → Closing. Add streaming output.

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21	Phase 4	Assignment 2 – Memory	Integrate vector DB. Store past debate outcomes. Use retrieval to improve arguments.
22	Phase 4	Assignment 2 – Polish	Test full debate. Verify agents respond to each other. Moderator declares winner.

Resource / Notes	Progress
Used github copilot for learning	Done
LangGraph intro video	Not Started
LangChain tool use docs	Not Started
LangGraph tutorial	Not Started
OpenAI / Anthropic tool docs	Not Started
LangChain structured output	Not Started
LangGraph error handling docs	Not Started
LangGraph tutorial	Not Started
LangGraph multi-agent guide	Not Started
LangGraph tutorial	Not Started
LangGraph supervisor example	Not Started
LangGraph docs	Not Started
	Not Started
	Not Started
	Not Started
LangChain RAG tutorial	Not Started
Chroma or FAISS docs	Not Started
LangGraph memory guide	Not Started
	Not Started
	Not Started

Resource / Notes	Progress
	Not Started
	Not Started

Notes	
Understood the basic difference How to use tools in Lang chain what is react Langchain's internal state machine graph Created few files related to each(agent, prompt,workflow) , using langchain framework & raw http calls , Integrated it with UI (Vue Js) pushed to --https://github.com/Escanor-Sensei/ai_agents_learning	

Notes





